

Human Computer Interaction

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Q1. [CLO-1]

Define binary image, grayscale image, and colored image. Explain how each image type is used in facial expression based interaction systems and discuss the trade-offs (e.g., storage, processing speed, feature extraction accuracy) associated with choosing one over the other in HCI applications.

Q2. [CLO-2]

You are evaluating a facial expression recognition system for drowsiness detection. During low-light stretching, the system suffers from high false negative rates and poor user trust. Explain how contrast stretching can improve image quality and address these usability issues.

Q3. [CLO-2]

You are evaluating a hand gesture recognition system for surgical display control. The system produces thick, imprecise hand boundaries leading to gesture misclassification. Explain how boundary extraction through morphological operations improves boundary precision and reduces cognitive load on surgeons.

Q4. [CLO-2]

You are evaluating a facial expression-based interaction system for an adaptive e-learning platform. The system detects holes in facial regions due to lighting reflections, causing incorrect engagement classification. Explain how region filling through morphological operations addresses this issue and improves system usability.

Q5. [CLO-3]

You are evaluating a facial expression recognition system used for driver drowsiness detection. The system captures grayscale facial images in real-time. However, during low-light conditions, user testing reveals:

- High false negative rate
- Increased response time
- Poor user trust

To improve system usability, you recommend applying Contrast Stretching. Given the following 5x5 grayscale image matrix:

35	38	40	42	45
36	39	180	43	44
37	40	185	41	46
38	41	44	43	47
39	42	45	44	48

8 bit image.

Q6. [CLO-3]

You are evaluating a hand gesture recognition system for contactless control of a surgical display screen. User testing reveals thick, imprecise boundaries leading to misclassification of gestures and high cognitive load on surgeons. You recommend applying boundary extraction through morphological operations. Given the following 7x7 binary image matrix (1 = hand region, 0 = background):

1	1	1	0	0	0	1
1	1	0	0	0	0	1
1	1	0	1	1	1	1
1	1	0	1	1	1	1
1	1	0	1	1	1	1
1	1	0	1	1	1	1
1	1	0	1	1	1	1

0 1 1 1 1 1 0
0 0 1 1 1 0 0

Structuring Element (3x3 square):

1	1	1
1	1	1
1	1	1

Q7. [CLO-3]

You are evaluating a facial expression-based interaction system for an adaptive e-learning platform. The system captures facial images to detect learner engagement. User testing reveals holes and gaps in detected facial regions due to lighting reflections, leading to incorrect engagement classification and learner frustration. You recommend applying region filling through morphological operations.

Given the following 8x8 binary image matrix (1 = facial region, 0 = background or hole) representing a segmented mouth region:

	1	2	3	4	5	6	7	8
1								
1	0	0	0	0	0	0	0	0
2	0	1	1	1	1	1	1	0
3	0	1	0	0	0	1	1	0
4	0	1	0	0	0	1	1	0
5	0	1	0	0	0	1	1	0
6	0	1	1	1	1	1	1	0
7	0	1	1	1	1	1	1	0
8	0	0	0	0	0	0	0	0

Structuring Element (3x3 cross-shaped):

0	1	0
1	1	1
0	1	0

Q8. [CLO-3]

You are designing a low-bandwidth telemedicine interface where rural healthcare workers receive medical images on low-end devices. The current system suffers from slow loading times and poor image clarity due to limited network bandwidth. You decide to apply **Bit Plane Slicing** to improve user experience by reducing data transmission while preserving critical diagnostic information. Given the following 4x4 grayscale image matrix (pixel values 0-255, 8-bit binary):

130	131	132	133
134	135	200	137
138	139	210	141
142	143	144	145